Lecture 14: Cayley–Hamilton Theorem

MATH203 Applied Linear Algebra

Annihilating Polynomials

How do you compute A^{100} ?

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Direct approach: multiply A by itself 100 times.

That is 99 matrix multiplications. **Very painful.**

Is there a shortcut?

What If We Knew a Shortcut for A^2 ?

Suppose someone told us:

$$A^2 = 5A + 2I$$

Then we can reduce **every** power of A:

$$A^3 = A \cdot A^2 = A(5A + 2I) = 5 \underbrace{A^2}_{5A+2I} + 2A$$

$$= 5(5A + 2I) + 2A = 27A + 10I$$

$$A^4 = A \cdot A^3 = A(27A + 10I) = 27 \underbrace{A^2}_{5A+2I} + 10A$$

$$= 27(5A + 2I) + 10A = 145A + 54I$$

Every power of A is just $\alpha A + \beta I$ — two numbers!

Why This Works

The relation $A^2 = 5A + 2I$ means:

$$A^2 - 5A - 2I = 0.$$

So the polynomial $f(t) = t^2 - 5t - 2$ satisfies $f(A) = 0$.

It **kills** the matrix A.

Definition

A polynomial $f(t) = c_0 + c_1 t + c_2 t^2 + \cdots + c_n t^n$ is called an **annihilating polynomial** of a matrix A if

$$f(A) = c_0 I + c_1 A + c_2 A^2 + \cdots + c_n A^n = 0.$$

If f annihilates A and has degree d, then every A^n can be written as a combination of $I, A, \ldots, A^{d-1}$.

Does Every Matrix Have an Annihilating Polynomial?

Consider a 2×2 matrix A.

The space $\mathbb{R}^{2\times 2}$ of all 2×2 matrices has dimension 4.

Now look at these **five** matrices:

$$I, \quad A, \quad A^2, \quad A^3, \quad A^4$$

Five vectors in a 4-dimensional space $\Longrightarrow$ **linearly dependent.**

So there exist coefficients c_0, c_1, c_2, c_3, c_4 (not all zero) with

$$c_0 I + c_1 A + c_2 A^2 + c_3 A^3 + c_4 A^4 = 0.$$

This is an annihilating polynomial of degree ≤ 4.

Existence in General

For an $n \times n$ matrix A:

$\mathbb{R}^{n \times n}$ has dimension n^2.

The $n^2 + 1$ matrices $I, A, A^2, \ldots, A^{n^2}$ must be **linearly dependent**.

$\implies$ an annihilating polynomial of degree $\leq n^2$ always exists.

matrix size	dimension of $\mathbb{R}^{n \times n}$	guaranteed degree
2×2	4	≤ 4
3×3	9	≤ 9
10×10	100	≤ 100

Problem: degree n^2 is big. And this argument does not tell us **how** to find the polynomial.

Can we do better?

Can we find an annihilating polynomial

of degree n (not n^2),

systematically?

The Characteristic Polynomial

Definition

For an $n \times n$ matrix A, the expression

$$\det(\lambda I - A)$$

is a polynomial in λ of degree n. It is called the **characteristic polynomial** of A.

Why is it a polynomial of degree n?

Each entry of $\lambda I - A$ is at most degree 1 in λ.

The determinant multiplies n entries (one per column),

so the result has degree $\leq n$.

The leading term comes from the diagonal:

$$(\lambda - a_{11})(\lambda - a_{22}) \cdots (\lambda - a_{nn}) = \lambda^n + \cdots$$

Example: Characteristic Polynomial of a 2×2 Matrix

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

$$\det(\lambda I - A) = \det \begin{pmatrix} \lambda - 1 & -2 \\ -3 & \lambda - 4 \end{pmatrix}$$

$$= (\lambda - 1)(\lambda - 4) - (-2)(-3)$$

$$= \lambda^2 - 5\lambda + 4 - 6$$

$$= \lambda^2 - 5\lambda - 2$$

This is a polynomial of degree 2 in λ.

The Cayley–Hamilton Theorem

Theorem (Cayley–Hamilton)

For any $n \times n$ matrix A, the characteristic polynomial $\det(\lambda I - A)$ is an **annihilating polynomial** of A.

That is: if $\det(\lambda I - A) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0$,

then $A^n + c_{n-1}A^{n-1} + \cdots + c_0 I = 0$.

This gives us an annihilating polynomial of degree exactly n,

computed **systematically** from $\det(\lambda I - A)$.

matrix size	dimension argument	Cayley–Hamilton
2×2	degree ≤ 4	degree $\leq$ **2**
3×3	degree ≤ 9	degree $\leq$ **3**
10×10	degree ≤ 100	degree $\leq$ **10**

Let's Check: Does It Really Work?

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \det(\lambda I - A) = \lambda^2 - 5\lambda - 2.$$

We need to verify: $A^2 - 5A - 2I = 0$.

$$A^2 = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix}$$

$$A^2 - 5A - 2I = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix} - \begin{pmatrix} 5 & 10 \\ 15 & 20 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

It works!

But **why** does it always work? We need a proof.

Proof Idea: The Adjugate

Recall the adjugate identity: for any square matrix M,

$$M^* \cdot M = \det(M) \cdot I.$$

Apply this to $M = \lambda I - A$:

$$\underbrace{(\lambda I - A)^*}_{\text{adjugate}} \cdot (\lambda I - A) \;=\; \det(\lambda I - A) \cdot I$$

Rearrange:

$$\frac{\det(\lambda I - A) \cdot I}{\lambda I - A} \;=\; (\lambda I - A)^*$$

The right side is a **polynomial** in λ (each entry of the adjugate is a determinant of a submatrix of $\lambda I - A$, hence a polynomial).

So $\det(\lambda I - A) \cdot I$ is **divisible** by $(\lambda I - A)$, with **zero remainder**!

From Divisibility to Cayley–Hamilton

We have:

$$\det(\lambda I - A) \cdot I = (\lambda I - A)^* \cdot (\lambda I - A) + \underbrace{0}_{\text{remainder}}$$

From polynomial division (which we will learn next):

> The **remainder** of dividing a polynomial by $(\lambda - A)$ equals the value of **plugging in** $\lambda = A$.

Since the remainder is 0:

$$\det(\lambda I - A)\Big|_{\lambda = A} = 0 \cdot I = 0$$

That is exactly the Cayley–Hamilton theorem!

$\rightarrow$ To make this rigorous, we need to understand **polynomial division**.

Polynomial Division

How do you **divide** polynomials?

The proof of Cayley–Hamilton needs polynomial division.

We start with ordinary (scalar) polynomials,

then extend to matrix polynomials.

Simplifying Fractions: Numbers

When the numerator is bigger than the denominator, extract the integer part:

$$\frac{7}{2} = \underbrace{3}_{\text{quotient}} + \frac{\overbrace{1}^{\text{remainder}}}{2}$$

$$\frac{13}{5} = \underbrace{2}_{\text{quotient}} + \frac{\overbrace{3}^{\text{remainder}}}{5}$$

The pattern:

$$\frac{\text{big number}}{\text{small number}} = \text{integer} + \frac{\text{remainder}}{\text{small number}}$$

Can we do the same thing for polynomials?

Simplifying Fractions: Polynomials

Consider the fraction

$$\frac{t^3+2}{t-1}.$$

The numerator t^3+2 has degree 3.

The denominator $t-1$ has degree 1.

Since degree of numerator $>$ degree of denominator,

we can extract a **polynomial quotient**:

$$\frac{t^3+2}{t-1} = \underbrace{(\cdots)}_{\substack{\text{polynomial} \\ \text{(degree 2)}}} + \frac{\text{remainder}}{t-1}$$

How do we find the quotient and remainder? $\rightarrow$ **Long division.**

Long Division: $(t^3 + 2) \div (t - 1)$

$$
\begin{array}{r|rrrr}
 & & t^2 & +t & +1 \\
\hline
t-1 & t^3 & & & +2 \\
 & t^3 & -t^2 & & \\
\cline{2-3}
 & & t^2 & & +2 \\
 & & t^2 & -t & \\
\cline{3-4}
 & & & t & +2 \\
 & & & t & -1 \\
\cline{4-5}
 & & & & 3
\end{array}
$$

Quotient: $t^2 + t + 1$

Remainder: 3

The Result

$$\frac{t^3+2}{t-1} = t^2+t+1+\frac{3}{t-1}$$

This is the polynomial analogue of $\frac{7}{2} = 3 + \frac{1}{2}$.

Now multiply both sides by $(t-1)$:

$$\boxed{t^3+2 = (t-1)(t^2+t+1)+3}$$

This is the **division formula**:

$$f(t) = \underbrace{(t-a)}_{\text{divisor}} \cdot \underbrace{Q(t)}_{\text{quotient}} + \underbrace{R}_{\text{remainder}}$$

The Remainder Is the Value at $t = 1$

$$t^3 + 2 = \underbrace{(t-1)(t^2+t+1)}_{\text{this part is 0 when } t=1} + 3$$

Plug in $t = 1$:

$$\underbrace{1^3 + 2}_{=3} = \underbrace{(1-1)}_{=0} \cdot (\cdots) + 3$$

$$3 = 0 + 3 \qquad \checkmark$$

The remainder 3 equals the value of $f(t) = t^3 + 2$ at $t = 1$.

Is this a coincidence?

Why Does This Always Work?

Dividing $f(t)$ by $(t - a)$ gives:

$$f(t) = (t - a) \cdot Q(t) + R$$

where R is a constant (degree of remainder $<$ degree of divisor).

Now plug in $t = a$:

$$f(a) = \underbrace{(a - a)}_{=0} \cdot Q(a) + R = R$$

So the remainder is **always** equal to $f(a)$.

$$\boxed{\text{Remainder of } f(t) \div (t - a) \;=\; f(a)}$$

Remainder Theorem

Theorem (Remainder Theorem)

Let $f(t)$ be any polynomial and let a be any number. Then

$$f(t) = (t - a) \cdot Q(t) + f(a)$$

for some polynomial $Q(t)$.

In words: the **remainder** of dividing $f(t)$ by $(t - a)$ equals $f(a)$.

Two ways to find the remainder:

1. **Long division** — divide step by step, read off the remainder.
2. **Direct substitution** — just compute $f(a)$. That *is* the remainder.

Example: $(t^2 - 5t - 2) \div (t - 1)$

Step 1. Predict the remainder by direct substitution:

$$f(1) = 1^2 - 5(1) - 2 = 1 - 5 - 2 = -6$$

Step 2. Verify by long division:

$$\begin{array}{r|rrr}
 & & t & -4 \\
\hline
t-1 & t^2 & -5t & -2 \\
 & t^2 & -t & \\
\cline{2-4}
 & & -4t & -2 \\
 & & -4t & +4 \\
\cline{3-4}
 & & & -6
\end{array}$$

$$\frac{t^2 - 5t - 2}{t - 1} = t - 4 + \frac{-6}{t - 1} \qquad \checkmark$$

Example: $(t^2 - 5t - 2) \div (t - 3)$

Step 1. Predict the remainder:

$$f(3) = 3^2 - 5(3) - 2 = 9 - 15 - 2 = -8$$

Step 2. Verify by long division:

$$\begin{array}{r|rrr}
 & & t & -2 \\
\hline
t-3 & t^2 & -5t & -2 \\
 & t^2 & -3t & \\
\cline{2-4}
 & & -2t & -2 \\
 & & -2t & +6 \\
\cline{3-4}
 & & & -8
\end{array}$$

$$\frac{t^2 - 5t - 2}{t - 3} = t - 2 + \frac{-8}{t - 3} \qquad \checkmark$$

When Is the Remainder Zero?

$$f(t) = (t-a) \cdot Q(t) + \underbrace{f(a)}_{\text{remainder}}$$

If $f(a) = 0$, the remainder vanishes:

$$f(t) = (t-a) \cdot Q(t)$$

So $(t-a)$ **divides** $f(t)$ with no remainder.

Corollary

$(t-a)$ divides $f(t)$ if and only if a is a root of f, i.e. $f(a) = 0$.

Example: $f(t) = t^2 - 5t + 6$. Then $f(2) = 4 - 10 + 6 = 0$,

so $(t-2)$ divides $f(t)$. Indeed $t^2 - 5t + 6 = (t-2)(t-3)$.

Back to Cayley–Hamilton

Recall our proof strategy:

1. The adjugate identity tells us

$$\det(\lambda I - A) \cdot I = (\lambda I - A)^* \cdot (\lambda I - A)$$

   so $\det(\lambda I - A) \cdot I$ is **divisible** by $(\lambda I - A)$.

2. The remainder of dividing by $(\lambda - A)$ equals the value at $\lambda = A$.

3. Since the division is exact (zero remainder), plugging in $\lambda = A$ gives 0.

To make this rigorous: we need polynomial division with **matrix coefficients**.

$\rightarrow$ **Next section.**

Dividing by $(\lambda - A)$

A Tempting but Wrong Proof

"Proof" of Cayley–Hamilton:

$$\det(\lambda I - A)\Big|_{\lambda = A} = \det(AI - A) = \det(0) = 0. \qquad \text{WRONG!}$$

The error: you **cannot** plug a matrix into $\det(\lambda I - A)$ directly.

The λ inside $\det(\lambda I - A)$ sits in a **scalar** slot.

Replacing it with a matrix changes the meaning of the expression entirely.

Why the False Proof Fails

Consider $\det(\lambda I_2) = \lambda^2$. The scalar polynomial is $f(\lambda) = \lambda^2$.

Plug in $B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$:

- **Correct**: expand first, then substitute:
  $f(B) = B^2 = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix}$ (a 2×2 matrix)
- Wrong: substitute inside the determinant:
  $\det(B \cdot I_2) = \det(B) = -2$ (a number, not a matrix!)

$\begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix} \neq -2$ They are not even the same type of object.

The Correct Approach

Step 1. Expand $\det(\lambda I - A)$ to get a **scalar** polynomial:

$$\det(\lambda I - A) = \lambda^2 - 5\lambda - 2$$

Step 2. Substitute $\lambda = A$ into the **expanded** polynomial:

$$f(A) = A^2 - 5A - 2I$$

Expand first, then substitute.

Never substitute a matrix directly into a determinant.

Dividing $\lambda^2 - 5\lambda - 2$ by $(\lambda - A)$

We treat A as a **constant** and divide:

$$\frac{\lambda^2 - 5\lambda - 2}{\lambda - A}$$

The divisor $(\lambda - A)$ has leading coefficient 1 (it is **monic**),

so long division works exactly as before.

The only difference: quotient and remainder now have **matrix coefficients**.

Long Division: $(\lambda^2 - 5\lambda - 2) \div (\lambda - A)$

$$
\begin{array}{r|lll}
 & & \lambda & +(A-5I) \\
\hline
\lambda - A & \lambda^2 & -5\lambda & -2I \\
 & \lambda^2 & -A\lambda & \\
\cline{2-3}
 & & (A-5I)\lambda & -2I \\
 & & (A-5I)\lambda & -(A^2-5A) \\
\cline{3-4}
 & & & A^2-5A-2I
\end{array}
$$

Quotient: $\lambda I + (A - 5I)$

Remainder: $A^2 - 5A - 2I$

The Result

$$\frac{\lambda^2 - 5\lambda - 2}{\lambda - A} = \lambda I + (A - 5I) + \frac{A^2 - 5A - 2I}{\lambda I - A}$$

Multiply both sides by $(\lambda I - A)$:

$$(\lambda^2 - 5\lambda - 2) \cdot I = [\lambda I + (A - 5I)] \cdot (\lambda I - A) + A^2 - 5A - 2I$$

The remainder $A^2 - 5A - 2I$ is exactly $f(A)$

— the value of plugging $\lambda = A$ into $f(\lambda) = \lambda^2 - 5\lambda - 2$.

Just like the scalar case!

Why the Remainder Equals $f(A)$

We have the division formula:

$$f(\lambda) \cdot I = Q(\lambda) \cdot (\lambda I - A) + R$$

where R is a constant matrix (does not depend on λ).

Plug in $\lambda = A$ into the **expanded** polynomial:

$$f(A) = Q(A) \cdot \underbrace{(AI - A)}_{=0} + R = R$$

So $R = f(A)$, exactly as in the scalar case.

Theorem (Matrix Remainder Theorem)

$$f(\lambda) \cdot I = Q(\lambda) \cdot (\lambda I - A) + f(A)$$

The remainder of dividing $f(\lambda) \cdot I$ by $(\lambda I - A)$ equals $f(A)$.

Existence and Uniqueness of Matrix Polynomial Division

For any **monic** divisor $d(\lambda)$ (leading coefficient $= I$),

the division

$$F(\lambda) = Q(\lambda) \cdot d(\lambda) + R(\lambda), \qquad \deg R < \deg d$$

always **exists** and is **unique**.

- **Existence**: the long division algorithm works because the leading coefficient I is invertible — each step reduces the degree by 1.
- **Uniqueness**: if two decompositions exist, their difference forces a degree contradiction.

See **Homework 4, Problem 9** for the full proof.

Proof of the Cayley–Hamilton Theorem

Now we have all the tools.

Let's **prove** Cayley–Hamilton.

The Adjugate Identity

Recall: for any square matrix M,

$$M^* \cdot M = \det(M) \cdot I$$

where M^* is the **adjugate** (matrix of cofactors, transposed).

Apply this to $M = \lambda I - A$:

$$(\lambda I - A)^* \cdot (\lambda I - A) = \det(\lambda I - A) \cdot I$$

Rearrange:

$$\frac{\det(\lambda I - A) \cdot I}{\lambda I - A} = (\lambda I - A)^*$$

The Adjugate Is a Matrix Polynomial

For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $\quad \lambda I - A = \begin{pmatrix} \lambda - 1 & -2 \\ -3 & \lambda - 4 \end{pmatrix}$

The adjugate (swap diagonal, negate off-diagonal):

$$(\lambda I - A)^* = \begin{pmatrix} \lambda - 4 & 2 \\ 3 & \lambda - 1 \end{pmatrix}$$

Write as a **polynomial in λ** with matrix coefficients:

$$(\lambda I - A)^* = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \lambda + \begin{pmatrix} -4 & 2 \\ 3 & -1 \end{pmatrix}$$

Each entry of $(\lambda I - A)^*$ is a cofactor = determinant of a submatrix of $\lambda I - A$, hence a **polynomial in λ**.

Verify: $(\lambda I - A)^* \cdot (\lambda I - A) = \det(\lambda I - A) \cdot I$

$$\begin{pmatrix} \lambda - 4 & 2 \\ 3 & \lambda - 1 \end{pmatrix} \begin{pmatrix} \lambda - 1 & -2 \\ -3 & \lambda - 4 \end{pmatrix}$$

$$= \begin{pmatrix} (\lambda-4)(\lambda-1) - 6 & -2(\lambda-4) + 2(\lambda-4) \\ 3(\lambda-1) - 3(\lambda-1) & -6 + (\lambda-1)(\lambda-4) \end{pmatrix}$$

$$= \begin{pmatrix} \lambda^2 - 5\lambda - 2 & 0 \\ 0 & \lambda^2 - 5\lambda - 2 \end{pmatrix} = (\lambda^2 - 5\lambda - 2) \cdot I \quad \checkmark$$

The adjugate $(\lambda I - A)^*$ is a **pure polynomial** — no $(\lambda I - A)^{-1}$ term.

Two Ways to Write the Same Fraction

Way 1 — polynomial division (from §3):

$$\frac{\det(\lambda I - A) \cdot I}{\lambda I - A} = Q(\lambda) + \frac{f(A)}{\lambda I - A}$$

where $f(A) = A^2 - 5A - 2I$ is the remainder.

Way 2 — adjugate identity (from §4):

$$\frac{\det(\lambda I - A) \cdot I}{\lambda I - A} = (\lambda I - A)^*$$

The right side is a **pure polynomial** — no $(\lambda I - A)^{-1}$ term!

Comparing the Two Decompositions

Setting them equal:

$$\underbrace{Q(\lambda)}_{\text{polynomial}} + \underbrace{\frac{f(A)}{\lambda I - A}}_{\text{not polynomial}} = \underbrace{(\lambda I - A)^*}_{\text{polynomial}}$$

Move the polynomial parts to one side:

$$\frac{f(A)}{\lambda I - A} = \underbrace{(\lambda I - A)^* - Q(\lambda)}_{\text{polynomial in } \lambda}$$

The left side has $(\lambda I - A)^{-1}$, which $\to 0$ as $\lambda \to \infty$.

A polynomial that $\to 0$ as $\lambda \to \infty$ must be **identically zero**.

Therefore $f(A) \cdot (\lambda I - A)^{-1} = 0$, so $f(A) = 0$. $\square$

Cayley–Hamilton Theorem — Proved!

Theorem (Cayley–Hamilton)

For any $n \times n$ matrix A:

if $\det(\lambda I - A) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0$,

then $A^n + c_{n-1}A^{n-1} + \cdots + c_0 I = 0$.

Proof summary:

1. Expand $\det(\lambda I - A) = f(\lambda)$, a scalar polynomial of degree n.
2. Polynomial division: $f(\lambda) \cdot I = Q(\lambda) \cdot (\lambda I - A) + f(A)$.
3. Adjugate identity: $f(\lambda) \cdot I = (\lambda I - A)^* \cdot (\lambda I - A)$.
4. Comparing: the adjugate decomposition has **zero remainder**.
5. By uniqueness: $f(A) = 0$. $\square$

Application: Computing Matrix Powers

For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, Cayley–Hamilton gives $A^2 = 5A + 2I$.

$$A^3 = A \cdot A^2 = A(5A + 2I) = 5\underbrace{A^2}_{5A+2I} + 2A = 27A + 10I$$

$$= 27\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + 10\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 37 & 54 \\ 81 & 118 \end{pmatrix}$$

$$A^4 = A \cdot A^3 = 27\underbrace{A^2}_{5A+2I} + 10A = 145A + 54I$$

$$= \begin{pmatrix} 199 & 290 \\ 435 & 634 \end{pmatrix}$$

Every power: $A^n = \alpha_n A + \beta_n I$ for just **two numbers** α_n, β_n.

Exercise

Let $B = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$.

1. Compute $\det(\lambda I - B)$.

2. Use the Cayley–Hamilton theorem to express B^2 in terms of B and I.

3. Compute B^5.